⚠ **TRAINING EXAMPLE — NOT A REAL IEC 62304 DELIVERABLE**

This project is a **training site, not a real medical device** under any regulatory definition. The device described in this document — "SentinelFlow 500" — is entirely **fictional**, invented for this course: nothing here is a genuine IEC 62304 deliverable, a real regulatory submission, or evidence of any real organisation's compliance with anything. It illustrates the *kind* of document a real SaMD/SiMD project would produce. See the document register for the full explanation.

Clause: 5.3 — Software Architectural Design · **Register:** Device Example Register

Document ID: SOP-04 · **Device:** SentinelFlow 500 (fictional) · **Safety class:** C
Status: SOP — template (Procedure not yet written for SentinelFlow 500)

> **Fictional worked example.** See SOP-01 for the disclaimer that applies to this whole document set, and the Device Example Register for the SentinelFlow 500 device description and what "template" status means here.

1. Purpose

This SOP defines how SentinelFlow 500's software requirements are transformed into a software architecture: the software items, their interfaces, the SOUP within the system, and the segregation needed for risk control.

2. Scope

Applies to the architectural design of SentinelFlow 500's control software (Class C — see SOP-01 §7.3). Covers Clause 5.3 in full — every sub-clause of 5.3 applies at Class B and C, so nothing in this SOP is specific to Class C except §5.3.5, segregation for risk control.

3. Responsibilities

ROLE	RESPONSIBILITY UNDER THIS SOP
Software Architect	Authors the architecture; identifies software items and their interfaces
Software Development Lead	Confirms the architecture allocates every software requirement to a software item
QA/RA	Confirms the architectural design review record meets §5.3.6
Risk Manager	Confirms the segregation described in §5.3.5 matches the risk control measures in SOP-11

4. Definitions & Abbreviations

TERM	MEANING
Software item	An identifiable part of a software system, per the architecture
SOUP	Software of Unknown Provenance — third-party or pre-existing software components
Segregation	Separation between software items intended to prevent one from compromising another's risk control function

5. References

- IEC 62304:2006+AMD1:2015, §5.3
- **SOP-03 — Software Requirements Specification**: the input this architecture is built from
- **SOP-05 — Software Detailed Design**: consumes each software item this architecture identifies
- **SOP-11 — Software Risk Management File**: source of the risk control measures §5.3.5 segregation exists to protect

6. What the standard requires

REF	TITLE	APPLIES AT CLASS C	WHAT MUST BE PRODUCED
5.3.1	Transform software requirements into an architecture	Yes	Documented architecture describing the software structure and identifying the software items
5.3.2	Develop an architecture for the interfaces of software items	Yes	Documented architecture for the interfaces between software items and between software items and external components
5.3.3	Specify functional and performance requirements of SOUP item	Yes	Functional and performance requirements specified for each SOUP item
5.3.4	Specify system hardware and software required by SOUP item	Yes	System hardware and software specified as necessary to support each SOUP item
5.3.5	Identify segregation necessary for risk control	Yes	Segregation between software items necessary for risk control, and a statement of how its effectiveness is ensured
5.3.6	Verify software architecture	Yes	Documented verification that the architecture implements the system and software requirements, supports the interfaces, and supports proper operation of any SOUP

(Quoted directly from `applicability.json` , identical in content to the self-referential set's 04. Note that 5.3.5 is the only requirement in this table that is Class C-specific in the standard itself — every other row here applies at Class B too.)

7. Procedure

Not yet written for SentinelFlow 500 — see the Device Example Register for what "template" status means. Once complete, this section would give SentinelFlow 500's actual software items — e.g. Motor Control, Sensor Monitoring, Alarm Management, Display/UI, Dose Validation — their interfaces, the SOUP they depend on, and the segregation between the dose-validation item and the rest of the system.

5.3.1–5.3.2 — The architecture and its interfaces

[Template.]

5.3.3–5.3.4 — SOUP requirements and environment

[Template.]

5.3.5 — Segregation necessary for risk control

[Template — see SOP-11 for the risk control measures this section would protect.]

5.3.6 — Verify software architecture

[Template.]

8. Records

RECORD	PRODUCED BY	RETAINED IN
Software architecture document, current version	Software Architect	Project technical file
Architectural design review record	QA/RA	Project technical file
Traceability matrix (requirements → software items)	Software Architect	Project technical file

9. Revision History

Illustrative only — SentinelFlow 500 has no real revision history.

VERSION	STATUS	DESCRIPTION
1.0	Template	Structure and requirements table complete; Procedure section pending